

SmartCab using DSA (Python)

Objective

Cab-booking platforms such as Uber, Ola, and Rapido use different pricing structures for different types of rides. The final fare may depend on the cab service, ride type, distance travelled, peak-hour charges, and discounts available to the customer.

The objective of this project is to develop a **Smart Cab Booking and Fare Comparison System using Object-Oriented Programming in Python.**

The system will allow a user to:

1. Enter their name and subscription status.
2. Select a pickup and destination location.
3. Enter the distance of the journey.
4. Select a cab service such as **Rapido, Uber, or Ola.**
5. Select a ride type such as **Bike, Auto, or Cab.**
6. Calculate the fare based on the selected service and ride type.
7. Apply a **20% peak-hour surcharge** when applicable.
8. Apply a **30% subscription discount** for subscribed users.
9. Compare the final fares offered by all three cab services.
10. Identify the cheapest available option.
11. Book the selected ride.

The project is designed to demonstrate the practical application of fundamental Object-Oriented Programming concepts, particularly **abstraction, inheritance, encapsulation, polymorphism, classes and objects, and magic methods.**

Fare Structure

Service Base Fare Rate/km

Rapido	₹30	₹15
Uber	₹50	₹20
Ola	₹40	₹18

Ride-Type Multipliers

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Fare Calculation

Basic Fare = Base Fare + (Rate per km × Distance)

Ride Fare = Basic Fare × Ride Type Multiplier

Peak Surcharge = 20% of Ride Fare (if peak hour)

Fare After Peak Charge = Ride Fare + Peak Surcharge

Subscription Discount = 30% (if subscribed)

Final Fare = Fare After Peak Charge - Subscription Discount